

Week 14 Quiz: Transformers, LLMs, and Foundational Models

Total points 20/20

A comprehensive quiz covering key concepts in Transformer architecture, LLMs, training techniques, and evaluation methods.

✓ The core limitation of vanilla RNNs for long-range dependencies is: * 1/1

- ☐ Too many parameters compared to attention
- ☒ Vanishing/exploding gradients through many sequential multiplications ✓
- ☐ Lack of a hidden state
- ☐ Inability to process character tokens

✓ In scaled dot-product attention, dividing by $\sqrt{d_k}$ primarily prevents: * 1/1

- ☐ Overfitting via regularization
- ☒ Softmax saturation from large dot products ✓
- ☐ Loss of positional information
- ☐ Gradient checkpointing overhead



✓ What is the main benefit of multi-head attention? *

1/1

- ☐ Eliminates the need for FFN layers
- ☒ Captures diverse interaction patterns in parallel
- ☐ Reduces complexity below linear time
- ☐ Removes the need for positional encodings



✓ In decoder self-attention for language modeling, the mask: *

1/1

- ☒ Prevents attending to future tokens
- ☐ Prevents attending to past tokens
- ☐ Forces uniform attention
- ☐ Disables attention on special tokens



✓ A benefit of sinusoidal (hardcoded) positional encodings is that they: *

1/1

- ☐ Require retraining to extend to longer sequences
- ☒ Enable length extrapolation without retraining
- ☐ Directly encode relative positions via learned per-head bias
- ☐ Remove the need for token embeddings



✓ Which architecture is bidirectional at pretraining time and well-suited for sentence encoding/classification? *1/1

- ☐ Decoder-only models (e.g., GPT)
- ☐ Encoder-decoder models
- ☒ Encoder-only models (e.g., BERT)
- ☐ Recurrent neural networks only



✓ Which approach is most commonly used today in LLMs for position information? *1/1

- ☒ RoPE applied inside attention
- ☐ Learned absolute positional embeddings added to tokens
- ☐ Sinusoidal PEs added to tokens only
- ☐ Relative position bias added to values only



✓ In causal self-attention, the masking is applied to: * 1/1

- ☒ The QKT score matrix before softmax
- ☐ The value matrix V after softmax
- ☐ The token embeddings before the attention block
- ☐ The output logits after the final linear layer



✓ The main purpose of a feedforward network sublayer between attention blocks is to: *1/1

- ☐ Aggregate sequence order
- ☐ Reduce the number of heads
- ☐ Replace positional encodings
- ☒ Provide token-wise nonlinearity and channel mixing



✓ The purpose of the KV cache at inference time is to: * 1/1

- ☒ Avoid recomputing past keys/values to reduce per-token latency
- ☐ Store gradients for backpropagation
- ☐ Save optimizer states between steps
- ☐ Increase parameter count without extra compute



✓ Compared to MHA, MQA/GQA reduces latency mainly because it: * 1/1

- ☐ Shares Q across heads
- ☐ Removes caching altogether
- ☒ Shares K/V across heads (or groups), shrinking the KV cache/bandwidth
- ☐ Doubles the number of heads



✓ In nucleus (top-p) sampling, the next token is sampled from: * 1/1

- ☐ The top-k tokens by probability
- ☐ All tokens above a fixed probability threshold τ
- ☒ The smallest set of tokens with cumulative probability mass $\geq p$
- ☐ The single highest-probability token



✓ Increasing softmax temperature T during decoding typically: * 1/1

- ☒ Flattens the distribution (more diversity)
- ☐ Sharpens the distribution (less diversity)
- ☐ Has no effect on probabilities
- ☐ Forces beam-search behavior



✓ The standard pretraining objective for decoder-only LLMs is: * 1/1

- ☐ Masked language modeling
- ☐ Next-sentence prediction
- ☒ Next-token prediction (autoregressive)
- ☐ Sequence autoencoding



✓ Supervised finetuning (SFT) is best summarized as: *

1/1

- ☐ Freezing the model and adding adapters only
- ☒ Collecting instruction/response pairs and tuning the model to change behavior ✓
- ☐ Training a reward model
- ☐ Doing reinforcement learning with PPO

✓ Low-Rank Adaptation (LoRA) for parameter-efficient finetuning works by: * 1/1

- ☐ Pruning attention heads
- ☐ Quantizing activations
- ☐ Training only embeddings
- ☒ Adding low-rank adapter matrices to the weight updates while keeping the base weights frozen ✓

✓ Mixed-precision training, as used in practice for LLMs, typically: *

1/1

- ☐ Stores all tensors in FP64 to avoid numerical error
- ☐ Performs the forward pass in FP64 and the backward pass in INT8
- ☐ Requires changing the model architecture
- ☒ Uses low precision for activations/gradients while keeping a high-precision copy of the weights for updates ✓



- ✓ What is a key limitation of standard Supervised Fine-Tuning (SFT) that Preference Tuning (e.g., RLHF) is specifically designed to address? *1/1
- ☐ SFT inherently prevents the model from understanding structural dependencies required for complex instruction following or formatting outputs.
 - ☒ SFT maximizes likelihood on good examples but lacks an explicit mechanism to penalize (provide negative signals for) undesirable/harmful outputs. ✓
 - ☐ SFT computationally demands an exponentially higher budget compared to Reinforcement Learning due to the need for labeled data.
 - ☐ SFT causes a catastrophic forgetting phenomenon that strictly prevents the model from generalizing to any prompts not seen during training.

- ✓ What is the core idea behind Chain of Thought (CoT)? * 1/1
- ☐ Forcing the model to output the final answer immediately to minimize token usage and latency.
 - ☐ Fine-tuning the model exclusively on high-fidelity encyclopedia articles to improve factual density.
 - ☒ Prompting the model to explain its thinking process before producing the final answer. ✓
 - ☐ Utilizing an external vector database retrieval system to fetch the answer directly from a corpus.



✓ LLM-as-a-judge typically involves: *

1/1

- ☒ Using a strong model (e.g., Gemini) to rate the outputs of other models based on a rubric. ✓
- ☐ Using a very small, distilled model to evaluate the capabilities of a much larger model.
- ☐ Asking the model to output a confidence score regarding its own generation.
- ☐ Utilizing a legacy BERT-based model specifically fine-tuned for binary sentiment analysis.

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